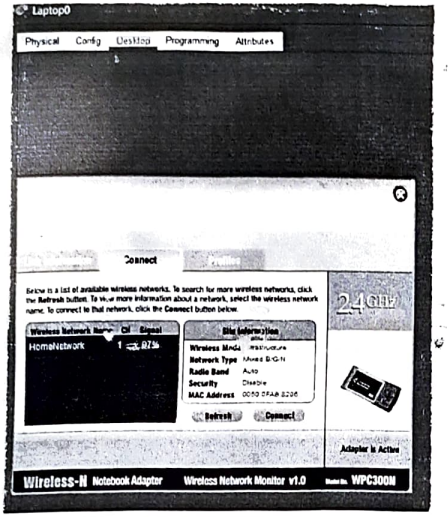
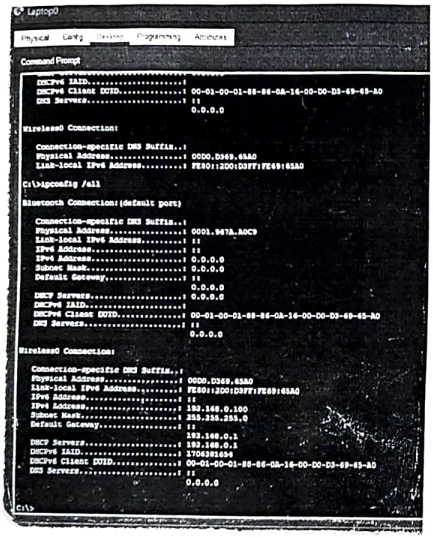
12/09/2026 EX:10 INTERNETWORKING WITH ROUTERS IN CISCO PACKET TRACER SIMULATOR.





(b) AIM:

To design and configure an internetwork using a wireless router, DHCP server and internet cloud in Cisco Packet Tracer, and verify connectivity between the PC and the cisco.com server. Trace simulate packet transmission between subnetworks without a router and observe the communication between the subnets.

PROCEDURE:

Step 1: Build the network topology.

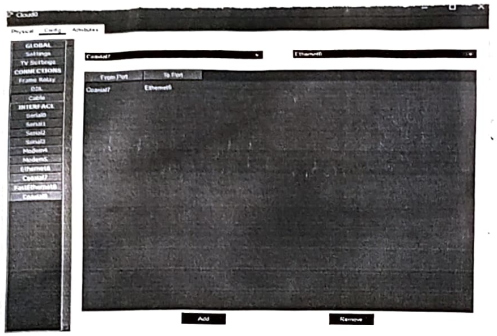
1. Open Cisco Packet Tracer.
2. Add the required devices.
   - ★ Wireless router.
   - ★ PC
   - ★ Laptop.
   - ★ Cable Modem
   - ★ Internet cloud.
   - ★ Cisco.com server.
3. Connect the devices according to the topology in the manual.
4. Use: ★ Copper Straight - through Cable between PC and wireless router
   - ★ Copper straight - through Cable b/w wireless router and Cable modem
   - ★ Coaxial Cable b/w Cable modem and Internet cable
   - ★ Copper straight through cable b/w Internet cloud and Cisco.com server.

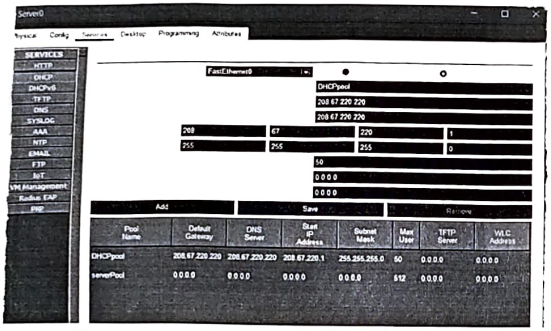
Step 2: Configure the Wireless Router.

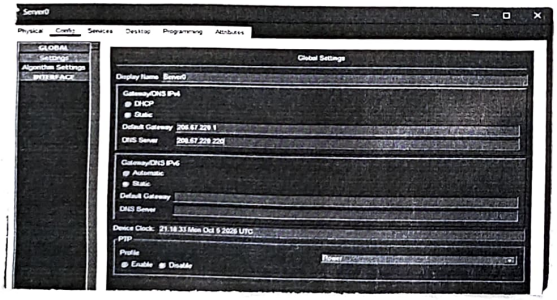
1. open the Wireless Router, select GUI → wireless tab.
2. Set the network name (SSID) to: HomeNetwork.
3. Go to the Setup tab, and verify the DHCP server is enabled.
4. Configure the DNS server address as: 208.67.220.220
5. Save the settings.

Step 3: Configure the Laptop.

1. Open the Laptop and go to the physical tab.
2. Power off the laptop. and remove the ethernet copper cable
3. Install the wireless WPC300N module and power on the laptop
4. Go to: Desktop → PC Wireless.
   - ↳ Select the Connect tab and Connect to the HomeNetwork Wireless Network.







Step 4: Configure the PC using DHCP.

1. open the PC.
2. select: Desktop → IP configuration.
3. Select the DHCP option.
4. The PC recieves its IPV4 configuration automatically from the wireless router
5. Open Cmd: ipconfig/all
6. verify that the PC recieves an IP address in the 192.168.0.x range.

Step 5: Configure the Internet cloud.

1. Open the Internet cloud and go to the physical tab.
2. Install the required cloud modules if they are not already present:
   - → PT-CLOUD-NM-1CX
   - → PT-CLOUD-NM-1CFE.
3. Go to config → cable and select. from Port: Coaxial To Port: Ethernet.
4. Click add and under Ethernet interface, select cable as the provider network.

Step 6: Configure Cisco.com Server as DHCP server.

1. open the Cisco.com Server.
2. select the services tab.
3. Select DHCP.
4. Turn the DHCP service On and configure the DHCP pool.

Step 7: Configure DNS Server.

↳ Select DNS under services and turn DNS on.

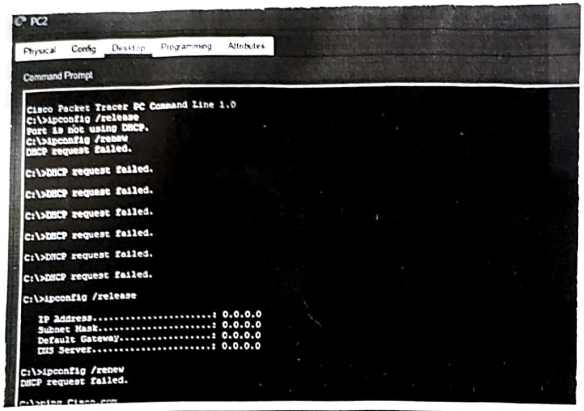
↳ Enter: Name: Cisco.com
Type: A record
Address: 208.67.220.220

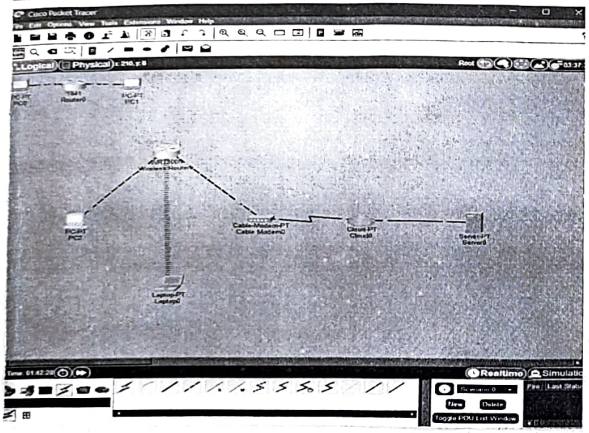
↳ click Add

Step 8: Configure Global settings.

↳ Select Config → settings and select static.

↳ Set: Gateway: 208.67.220.1
DNS Server: 208.67.220.220.





Step 9: Configure Fast Ethernet 0.

↳ Select Config → Fast Ethernet 0.
↳ Select Static.
↳ Enter: IP Address: 208. 67. 220. 220
Subnet Mask: 255. 255. 255. 0.

Step 10: Configure Fast Ethernet 0.

↳ Select Config → FastEthernet 0 and select static.
↳ Enter: IP Address: 208. 67. 220. 220.
Subnet Mask: 255. 255. 255. 0.

Step 11: Verify Connectivity.

↳ Refresh PC IPV4 settings.
* Open the PC.
* Go to Desktop → Cmd Prompt.
* Enter:
ipconfig/ release.
ipconfig/ renew
ipconfig/ all.

Step 12: Test Cisco.com Connectivity.

* In Cmd enter:
* ping cisco.com.
↳ wait for the response.
↳ Verify that 4 replies are received.

Step 13: Save the file.

↳ Save the Packet Tracer file using file → Save As.
↳ Give it a suitable name such as practical_10B.
↳ Save the file
↳ Close

RESULT:

Thus, an internetwork using a wireless router, DHCP server and Internet cloud was successfully designed and configured in Cisco Packet Tracer, and connectivity was successfully verify.